

Homework Set #5 – Due 11:59pm, Friday, May 22.
MATH 447 SPRING 2026

Note:

- This homework is based on Chapter 10 Part 1 & Part 2.
- For your convenience, all the data files and the .tex file for this problem are included.
- You need to submit two files: A PDF file which contains all the answers and an R script file (with an “.R” extension) containing all the code. Any answers presented in the R script file only will not be graded.
- Your answers must be presented in the same order as the problems given in this assignment, including all graphs and R outputs, if applicable (i.e., the graphs and outputs must be in the same order and must not be relegated to the back of your assignment).
- Use your judgment to include an appropriate amount of R code and output in your answers as necessary. For example, it is not necessary for the instructor to see every single observation of the dataset you used.
- All the R code must be well documented and submitted as a single R script file (which has the “.R” extension) to Canvas.
- If you need an extension, you need to ask for permission before 5pm on the day the homework is due.

The grading scheme is as follows:

- This homework assignment will be graded out of 30 points.
- Each problem will be graded out of 10 points.
- If the R code in the .R file is missing or incomplete, at most 5 points will be deducted.

Textbook Problems: Refer to *An Introduction to Statistical Learning with Applications in R*, First Edition, by James, Witten, Hastie, and Tibshirani.

1. Principal component analysis (PCA). Use HW5code.R to answer these questions.
 - (a) Present a biplot and provide a reasonable interpretation for ϕ_1 and ϕ_2 (the first and second principal component loading vector, respectively). It may be helpful to examine the actual values of the loading vectors to understand what they represent.
 - (b) By referring to both the first and second principal component, discuss important characteristics of Observation #6083.
 - (c) Present a scree plot and plot of the cumulative proportion of variance explained, and decide how many principal components should be used. Be sure to justify your choice.
2. PCA and K -means clustering. When answering these questions, modify HW5code.R appropriately.
 - (a) Interpret ϕ_1 , ϕ_2 , and ϕ_3 (the first, second, and third principal component loading vector, respectively). It may be helpful to examine the actual values of the loading vectors to understand what they represent.
 - (b) Perform K -means clustering on (Z_1, Z_2, Z_3) using $K = 2$ and $K = 3$. Then, plot the results in three scatterplots (Z_1, Z_2) , (Z_1, Z_3) , and (Z_2, Z_3) separately, and present them in your answer. Each cluster should be represented using different colors. Moreover, present the K clusters as a 3D scatterplot as well. In total, you should present $3 \times 2 + 2 = 8$ plots. Then, justify whether $K = 2$ or $K = 3$ is more reasonable.
 - (c) Using $K = 3$, interpret each cluster.
3. # 10.7.3. For (a), use `plot()`. Skip (b). Use `label <- c(2,1,1,2,1,2)` for (c).